

UniAdapFuse

Reliability-Aware Multimodal Fusion for UAV Disaster Perception

Project overview · what we did, how we did it, what we found · thesis-defense questions with model answers

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How to use this document

Part A (sections 1–9) is the full overview: problem, data, method, every result table and the demo app.

Part B holds 65 likely defense questions with model answers, grouped by topic; a few flag phrases to avoid. **Part C** is a one-page sheet of numbers to memorise and wording to use.

Read section 2 first. It lists problems found while preparing this guide that a panel could catch, including one factual mismatch between the thesis text and the D-Fire data on disk.

Sources: thesis chapters 1–9 (LaTeX source as of 2 October 2026), the saved training and evaluation outputs referenced there, *demo_app/eval/*.md*, and on-disk checks made while writing this guide.

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Part A · Project overview

1. Executive summary

UAVs in disaster response must spot hazards and people even when one sensor is blinded by smoke, darkness or rotor noise. Most aerial fusion models weight their sensors with fixed weights. We built **UniAdapFuse**: a framework that encodes RGB, thermal and audio separately, scales each modality's token with a learned **Reliability and Uncertainty Estimator (RUE)**, mixes the tokens with cross-modal attention, and classifies the disaster type and victim presence. It keeps running when any sensor is missing. We also tested a physics-guided **Hazard Prior Gate (HPG)** for smoke and fire detection, a joint classification-plus-detection graph, and a distilled student model.

We trained **26 classification configurations** on a **72,997-row corpus** assembled from six public datasets, **7 detectors** on the D-Fire smoke/fire benchmark, 6 unimodal backbone variants, 6 public-protocol runs (AIDER, FLAME) and a grouped-split retraining. We then audited what the data can support. We also built a demo app at the supervisor's request.

98.74% AdapFuse-v1 combined macro-F1 (row-level split)	98.85% RGB-only baseline: fusion does not beat it	89.14% AdapFuse-v1 on a capture-group split (−9.60 pp)	99.24% Best model: AdapFuse-v1 + EfficientNet-B0 (row-level)
77.10% Best D-Fire detector mAP ₅₀ (YOLO26s + CLAHE)	−0.94 pp HPG vs baseline YOLO26n; also 1.9× slower to train	53.7→99.3% Joint model disaster accuracy: naive → revised recipe (53.71 → 99.26)	0.0 Joint model detection mAP ₅₀ : unified detection not achieved

The honest verdict in five sentences

1. The architecture works as software: one model fuses any subset of three inputs, runs at 14.32 ms on a workstation GPU, and classifies scenes at near-ceiling accuracy on the row-level split.
2. Fusion gives no accuracy gain, because the public data do not contain independent sensors: the thermal stream is computed from RGB, and the ESC-50 audio is paired by label, which also leaks the fire/clean label.
3. The learned reliability gates saturate at 0 or 1 and do not react to corruption, so the RUE is not yet a dynamic reliability estimator; the nuisance head is never supervised.
4. The near-ceiling scores are partly leakage and dataset identity: a capture-group split costs 9.60 points, and victim presence is fully determined by the source dataset.
5. The HPG and the joint detector are negative results. The main lasting contribution is a measured statement of what data a credible UAV fusion evaluation needs: synchronised sensors, degradation labels and source-decorrelated splits.

2. Fix or prepare before the defense

These issues were found while preparing this guide, by comparing the thesis text with the code and data on disk. The first one is a factual mismatch that a reader who knows D-Fire would notice.

#	Issue	Evidence	Suggested fix	Risk
1	D-Fire split is described wrongly. Chapter 2 (D-Fire paragraph) and Chapter 5 (Tables 5.1, 5.2 and their text) say D-Fire was re-split 70/20/10 with a 2,153-image test set, that "all published methods were re-evaluated on our test set", and that the re-split avoids the near-duplicate contamination reported by Ruangsang and Pramkeaw.	<i>adaptfuse_od/data/datasets/dfire</i> : train 14,122 + val 3,099 = 17,221 (the official training set) and test 4,306 (the official test set). The 77.10% figure matches the 4,306-image test folder (re-verified at 0.7711 during the demo work). Published rows are values copied from the papers; they were not re-run.	State the official split (17,221 / 4,306) and our 82/18 train/val carve-out; say published values are as reported on the same official test set; delete the "re-evaluated" and "avoids contamination" sentences; note that the 46% near-duplicate finding applies to our test set too.	Critical
2	Design B (early fusion) is trained from scratch , but this is not disclosed.	<i>models/full_model.py</i> :193 <i>resnet18(weights=None)</i> with a new 5-channel conv1. All other designs use ImageNet weights; Chapter 4 states every visual encoder has ImageNet weights.	Add one sentence to Table 4.3 and Chapter 5: the B vs D/E gap mixes fusion topology with the lack of pretraining.	High
3	RT-DETRv2-R18 and YOLO11n D-Fire results have no logs in the repo.	<i>outputs/benchmark_rtdetr_r18/</i> is empty; <i>benchmark_yolov11n/</i> holds only config.yaml.	Copy the logs and checkpoints you have into <i>adaptfuse_od/outputs/</i> and bring them to the defense.	Medium
4	HPG location is worded two ways.	Chapter 1 objective: HPG "injects ... into the detection backbone at stride 4". Chapter 4: a residual correction of the input before the stem, with cues pooled to stride 4.	Use the Chapter 4 wording everywhere.	Low
5	Figure 1.1 says "67k" and has a garbled label.	The caption already notes the correct 70,997.	Regenerate the figure if possible; otherwise be ready to point to the caption.	Low
6	Demo calibration stage was selected on test.	<i>demo_app/eval/classifier_eval.md</i> : "selected_by: adoption rule on ... test.csv".	If you show demo accuracy, call it optimistic or re-select the stage on validation.	Low

3. Problem, motivation and objectives

Satellites revisit every 5–16 days and are blocked by cloud; ground sensors cover only points. UAVs deploy fast with sub-metre resolution, but each on-board sensor fails in its own way: RGB under smoke, darkness and glare; thermal at low resolution (typically 320×240) and with false alarms on sun-heated roofs and soil; audio under rotor noise. Fusion could cover each failure with another sensor, but existing aerial fusion models face three barriers: compute beyond UAV budgets, fixed modality weights that cannot down-weight a blinded camera, and negative transfer when dense box regression is trained together with scene classification.

3.1 Four challenges the thesis targets

- **Scale variance:** smoke plumes span hundreds of pixels while distant survivors are under 16×16 px.
- **Sensor degradation and false alarms:** the model should estimate per-modality reliability and gate inputs accordingly.
- **Multi-task gradient conflict:** dense detection gradients can damage a shared classification backbone.
- **Edge deployment:** targets of ≤500 ms (ideally <20 ms) latency inside a 10–15 W power envelope.

3.2 Objectives and what actually happened

Objective	Outcome
Hazard Prior Gate for smoke/fire detection at low cost	Not achieved: -0.94 pp mAP_{50} and $\sim 1.9\times$ longer training (8.65 h vs 4.45 h); validation curve tracks the baseline epoch for epoch.
RUE-driven Top-Down Context Gate for dense detection	Implemented. Gates stay nearly constant under corruption; the joint detector it conditions is not validated.
Phased curriculum + stop-gradient to stop negative transfer	Classification recovered (98.57% combined F1), but the same run also adds distillation, lower lr, longer training and reweighting; detection $mAP_{50} = 0.0$.
Benchmark modern detectors on D-Fire; test missing-modality behaviour on 72,997 samples	Done with independently trained detectors; missing modalities tested via two-sensor subsets and dropout; fusion (98.74%) does not exceed RGB-only (98.85%).
Compress via knowledge distillation (AdapFuse-v2)	Only 4.9% fewer parameters (12.94M $\rightarrow$ 12.30M) because backbones are unchanged; no embedded profiling.

3.3 Claimed contributions (as worded in the thesis)

- **UniAdapFuse**: RUE-gated multimodal fusion before cross-modal attention; one model handles any sensor subset.
- **Hazard Prior Gate**: physics cues (flame chromaticity, smoke achromaticity, edges) as a bounded input correction for YOLO26n, reported in full as a negative result.
- **Revised joint-training recipe** that reverses the classification collapse (53.71% $\rightarrow$ 99.26% disaster accuracy), with the finding that the joint detection branch does not learn.
- **Benchmark and audit** of 26 classification configurations and 7 detectors, showing that on current public data fusion does not beat RGB-only, a grouped split costs 9.60 points, label-matched audio leaks the label, and source identity determines the victim label.

4. How we did it: the workflow

1. Data curation	2. Design	3. Training	4. Evaluation	5. Audit	6. Demo app
6 public sources $\rightarrow$ 72,997 rows, 2 labels, 3 presence flags; balancing; row-level and grouped splits	6 fusion designs (A1-F), backbone selection by budget, joint model, HPG, KD student	Same pipeline for every model: AdamW, focal loss, AMP, 20% modality dropout; RTX 5090 32 GB	Row-level test, grouped split, AIDER and FLAME protocols, D-Fire, corruption sweeps, ablations, latency	Provenance counts, audio-leak analysis, per-source decomposition, log checks (HPG timing, NaN losses)	Live and batch web apps on one shared engine; calibrated ensemble + detectors

Every model, including the baselines and the backbone variants, was trained through the same AdapFuse-UAV pipeline (same loader, balancing, augmentation, focal loss, mixed precision, early stopping and evaluation), so differences come from the architecture or recipe changes we list, not from different codebases. About 390 GPU-hours of training are recorded in the saved logs (≈ 280 classification, ≈ 110 detection).

5. Data

5.1 Sources

Source	Scenario	Rows	Notes
FLAME	Wildfire (UAV)	47,992	Largest source (65.7%). Thermal entries in our metadata are RGB-derived pseudo-thermal.
C2A	Collapse, crowd, victim	10,215	Synthetic composites: human poses pasted on UAV disaster scenes; every row victim-positive.
AIDER	Fire, flood, collapse, traffic, normal	6,433	Aerial disaster scene classification benchmark.
SARD	Person visible	5,755	Aerial search-and-rescue; every row victim = 1 and disaster = clean.
ESC-50	Audio events	2,000	Clips reused across 39,003 rows; label-matched, not synchronised.
FireNet	Fire / no-fire	602	Video frames; in the grouped test every row is fire.

Source	Scenario	Rows	Notes
Total		72,997	Row-level split: 51,097 train / 10,949 val / 10,951 test (stratified by disaster label).

Labels. Disaster: 0 clean, 1 fire/smoke, 2 collapse/flood, 3 other hazard. Victim: 0 no person visible, 1 person visible. Each row also has presence flags for RGB, pseudo-thermal and audio; missing inputs are zero-filled.

5.2 What the modalities really are

Modality	What is in the metadata	Consequence
RGB	70,997 measured images	The only independent sensor.
"Thermal"	70,997 pseudo-thermal maps, all computed from the RGB frame; no sensor-IR path	Cannot see anything RGB cannot; degrades together with RGB; not a night-time fallback.
Audio	39,003 assignments drawing on 2,000 ESC-50 clips; only clean and fire rows have audio; 52.8% of test rows have audio	Every one of the 20,433 fire rows with audio gets one of the 40 "crackling fire" clips; all 18,570 clean rows with audio get one of the other 1,960. The clip class encodes the fire/clean label .

Key point for the defense. The "multimodal" fusion evaluated here is RGB + an RGB-derived map + label-leaking audio. It is not synchronised tri-modal sensing. This single fact explains why fusion cannot beat RGB-only.

5.3 Class balancing

Training split	Raw	Full balanced	Stratified balanced (used)
Total rows	51,097	98,556	49,276
Clean / Fire / Collapse / Other	41.0 / 46.5 / 8.5 / 4.0%	25% each	12,319 each (25%)
Victim present	21.7%	50.0%	50.0%
Synthetic (duplicated) rows	0%	48.2%	48.1%
Validation	10,949	21,030	10,516
Test	10,951, natural distribution, never resampled		

Stage 1 oversamples every disaster class to the majority count (23,746); stage 2 balances victim presence 50/50. For compute, a source-stratified 50% sample (minimum 5 rows per source) gives the 49,276-row training set used by all 26 classification models. Duplicated rows get stronger augmentation. Because the test set stays imbalanced, **macro-F1** is the primary metric. **Combined F1** = $\frac{1}{2}$ (disaster macro-F1 + victim macro-F1).

5.4 Two ways of splitting

	Row-level split (main results)	Capture-group split (audit)
Rows (train / val / test)	51,097 / 10,949 / 10,951	50,845 / 11,055 / 11,097
Grouping	None: adjacent video frames can sit on both sides	9,267 / 2,003 / 2,002 disjoint groups; FLAME in blocks of 250 frames; C2A and SARD by original image; ESC-50 folds 1-3 / 4 / 5
What it measures	Within-collection performance	A more conservative estimate on unseen capture sequences

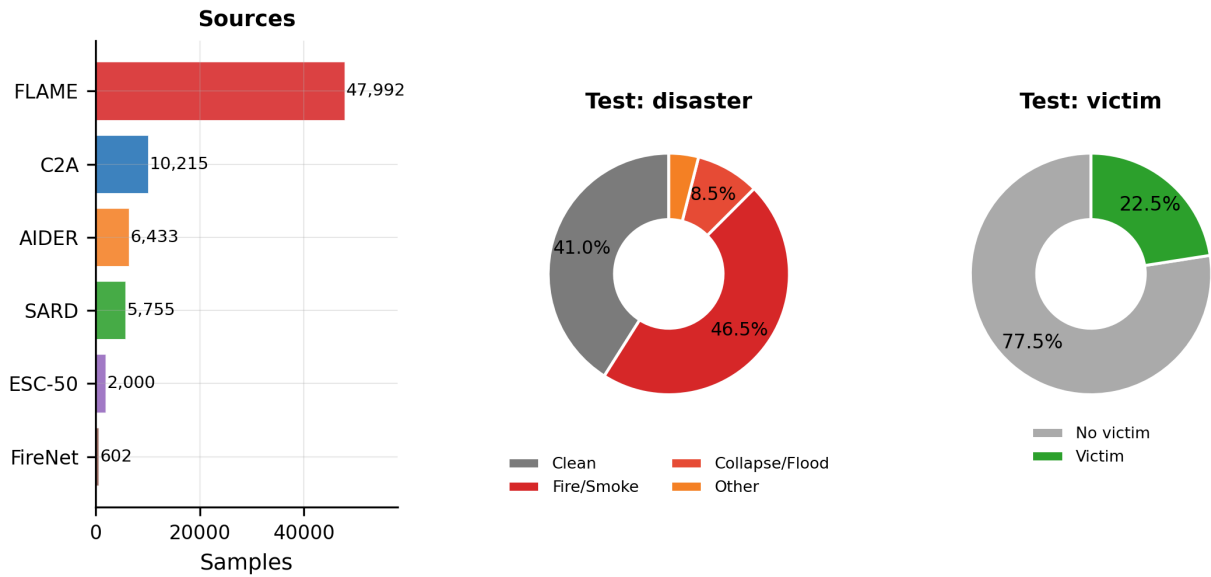


Figure 1 (thesis Fig. 4.3). Source mix and test-set label distribution. FLAME dominates with 65.7% of rows.

6. Method

6.1 AdapFuse-v1, the proposed classifier

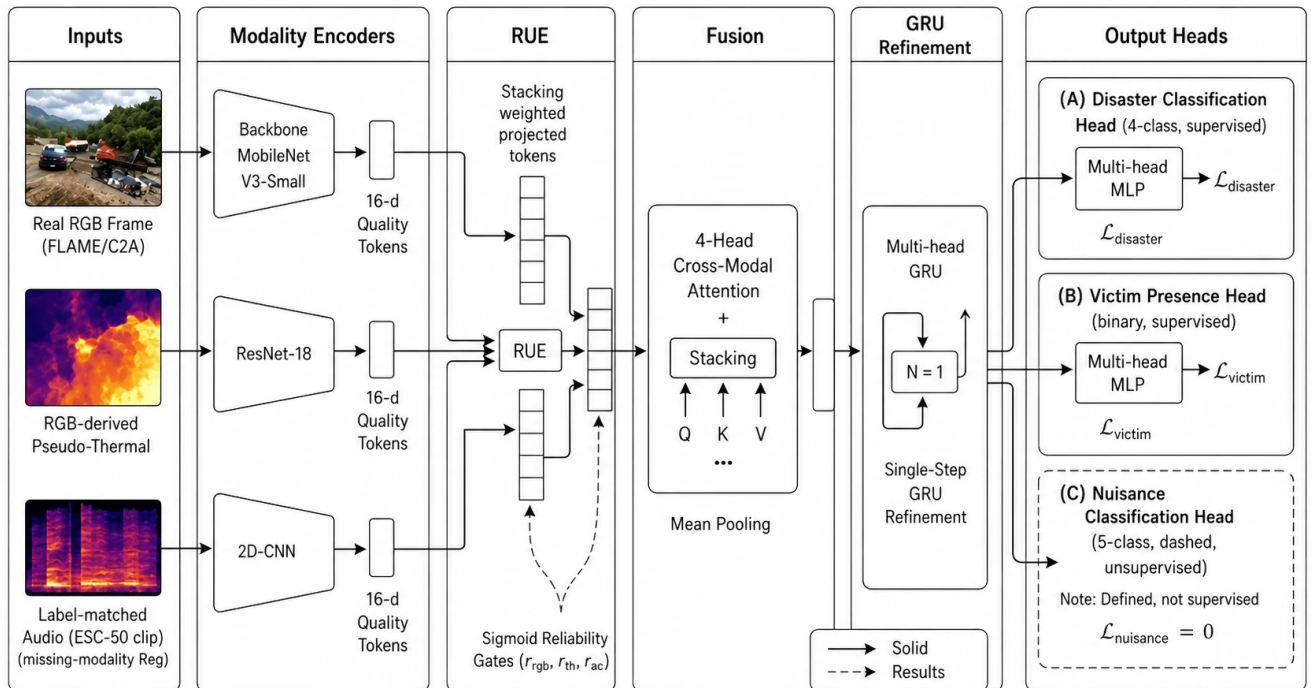


Figure 2 (thesis Fig. 4.1). AdapFuse-v1 graph. Three encoders produce 192-d tokens and 16-d quality tokens; the RUE gates the tokens; 4-head attention and mean pooling fuse them; a single-step GRU refines; disaster and victim heads are supervised, the nuisance head is not.

- **Encoders.** RGB: MobileNetV3-Small (ImageNet, 576-d $\rightarrow$ 192). Pseudo-thermal: ResNet-18 with one input channel (pretrained first-layer weights summed). Audio: 3-layer 2D-CNN (32 $\rightarrow$ 64 $\rightarrow$ 128) on a 64 $\times$ 63 log-mel spectrogram of a 2 s, 16 kHz clip.
- **RUE.** A 2-layer MLP over the three 16-d quality tokens and 6 feature statistics outputs gates $r_i = \sigma(W_2 \text{ReLU}(W_1[q_{\text{rgb}} \| q_{\text{th}} \| q_{\text{au}}] + b_1) + b_2) \in [0, 1]$. Each token is multiplied by its gate; a missing modality's gate is forced to 0.

- **Fusion.** 4-head self-attention over the three gated tokens, mean pooling, then a GRU (hidden 192) called with sequence length 1, so it acts as a gated refinement layer, not temporal memory.
- **Heads.** Disaster (4 classes) and victim presence (binary) are supervised. A 5-class nuisance head (clean, solar heating, RGB false fire, audio false alarm, occlusion) is defined, but the trainer passes no nuisance labels, so its loss is zero.
- **Loss.** $L = w_d \cdot FL_{\text{disaster}} + w_v \cdot FL_{\text{victim}} - \lambda_r \cdot \text{Var}(r)$, with focal loss $\alpha = 0.25$, $\gamma = 2$.
- **Size and speed.** 12.94M parameters; 14.32 ms end-to-end at batch 1 on an RTX 5090 workstation.

6.2 The six fusion designs

Design	Approach	Main limitation observed
A1 RGB-only	MobileNetV3-Small → FC	No thermal or audio cue
A2 Pseudo-thermal-only	ResNet-18 (1-ch) → FC	No colour; over the compute budget
A3 Audio-only	2D-CNN on log-mel	Label-matched ESC-50; 47% of test rows silent
B Early fusion	5-channel input into one ResNet-18	Loses modality-specific features; ResNet-18 trained from scratch in code (see fix list)
C Late fusion	Three models, learned weighted average of probabilities	No shared intermediate features
D Intermediate, fixed	Per-sensor backbone → 192-d → concatenate → MLP	All sensors weighted equally
E AdapFuse-v1 (ours)	D + RUE gating + cross-modal attention + nuisance head	Gates saturated; GRU single-step; nuisance unsupervised; +0.05 pp over D
F AdapFuse-v2 (ours, KD)	E as teacher → student with projection 96 (half)	Only 4.9% fewer parameters; no on-device latency

6.3 Backbone selection

Criteria: public pretrained weights; a per-stream budget of about 1 GMAC and 5M parameters at 224×224 (three encoders run per frame); highest ImageNet top-1 inside the budget; and three different design families. Parameters, MACs and training memory were measured for 13 candidates with one script.

Backbone	Params (M)	GMACs	Top-1	Decision
MobileNetV3-Small	0.93	0.06	67.7	Selected (default RGB)
MobileViT-XXS	0.95	0.30 (lower bound)	69.0	Selected (only in-budget hybrid)
EfficientNet-B0	4.01	0.39	77.7	Selected (best top-1 in budget)
ResNet-18	11.18	1.81	69.8	Kept only as the A2 thermal baseline (over budget)
ResNet-50, ConvNeXt-T, Swin-T, ViT-B/16	23–86	4.1–17.6	76–82.5	Rejected: 4–18× the budget for ≤4.8 top-1 points

Audio encoders: AudioCNN (0.09M), AudioCNN + GDBlock (dilations 1/3/5, 0.06M), and a CNN6-style encoder (1.55M). The CNN6-style encoder was meant to load PANNs AudioSet weights, but it uses 3×3 kernels while the checkpoint has 5×5, so only 8 of 20 tensors (batch-norm only) load: it is effectively trained from scratch.

6.4 UniAdapFuse: the joint classification + detection graph

- Feature pyramid (C2–C5) from the RGB and thermal backbones, merged with lateral connections.
- **Top-Down Context Gate (TDCG):** the fused scene vector and RUE scores modulate detection features with FiLM.
- **Bottom-Up Spatial Guidance (BUSG):** pooled box saliency maps feed the victim classifier.
- **v2 recipe:** 15-epoch classification-only warm-up; *.detach()* on pyramid features entering the detection neck; learning rates 0.1× backbone, 1× classifier, 0.3× detection; distillation from AdapFuse-v1; base lr 5×10^{-4} ; 80 epochs (patience 25); loss $2.0 \cdot L_{\text{dis}} + 0.5 \cdot L_{\text{vic}} + 0.2 \cdot L_{\text{det}} + 0.75 \cdot L_{\text{KD}}$.

6.5 Hazard Prior Gate (HPG)

Placed in front of the YOLO26n stem. Cue maps from the input frame: $P_{\text{fire}} = \text{ReLU}(R - \frac{1}{2}(G + B))$, $P_{\text{smoke}} = 1 - (\max - \min)/(\max + \epsilon)$ over RGB, and a Sobel edge map of the luminance. Cues and image are pooled to quarter resolution, encoded by a small CNN and upsampled into an RGB residual Δx and a spatial gate g ; output = $x + s \cdot g \cdot \Delta x$ with learned strength $s \leq 0.25$.

6.6 Training configuration

Setting	Value
Hardware	NVIDIA RTX 5090 (32 GB) for training and benchmarking; not the embedded target
Optimiser	AdamW (β 0.9/0.999, weight decay 10^{-4}), lr 10^{-3} (5×10^{-4} for UniAdapFuse v2)
Schedule	5-epoch linear warm-up, cosine to 10^{-5} . Logged lr instead cycles every ~ 8 epochs for C, D, E, F, domain prompts and naive UniAdapFuse
Batch / epochs	16 (8 for EfficientNet-B0); up to 50 epochs (80 for UniAdapFuse v2), early-stopping patience 15-25; AMP fp16
Data	Stratified balanced subset (49,276 rows); 20% random single-modality dropout
Corruption augmentation	Not active: corruption_prob = 0.3 is set but no corruption_type is passed; corruptions are used only in offline stress tests
Distillation	Logit KD ($T = 4.0$) + feature KD ($\lambda = 0.5$); used by AdapFuse-v2 and UniAdapFuse v2

7. Results

7.1 Smoke and fire detection on D-Fire

Model	Time (h)	Prec.	Rec.	mAP ₅₀	mAP _{50:95}	AP _{smoke}	AP _{fire}
YOLO26s (CLAHE)	42.6	0.769	0.701	77.10	44.80	84.85	69.36
RT-DETRv2-R18	7.85	0.742	0.678	73.80	42.15	81.20	66.40
YOLO26n (CLAHE, fine-tune)	+4.28	0.736	0.666	72.58	40.88	79.64	65.51
YOLO11n	4.60	0.728	0.659	72.25	40.52	79.10	65.40
YOLO26n (baseline)	4.45	0.712	0.647	71.47	40.05	78.33	64.60
Hazard-YOLO26n (HPG)	8.65	0.715	0.636	70.53	39.89	78.12	62.94
YOLOv10n	5.07	0.681	0.619	67.75	37.51	75.28	60.23

Recipes differ: YOLO26n, HPG and YOLOv10n share 512 px, AdamW, 40 epochs, batch 16; YOLO26s uses 640 px, SGD, 100 epochs, batch 4; the YOLO26n CLAHE run continues the YOLO26n checkpoint for 20 more epochs at 640 px. Smoke AP exceeds fire AP by 13.7–15.5 points in every model. Test set: the official D-Fire test folder (4,306 images, see fix list item 1).

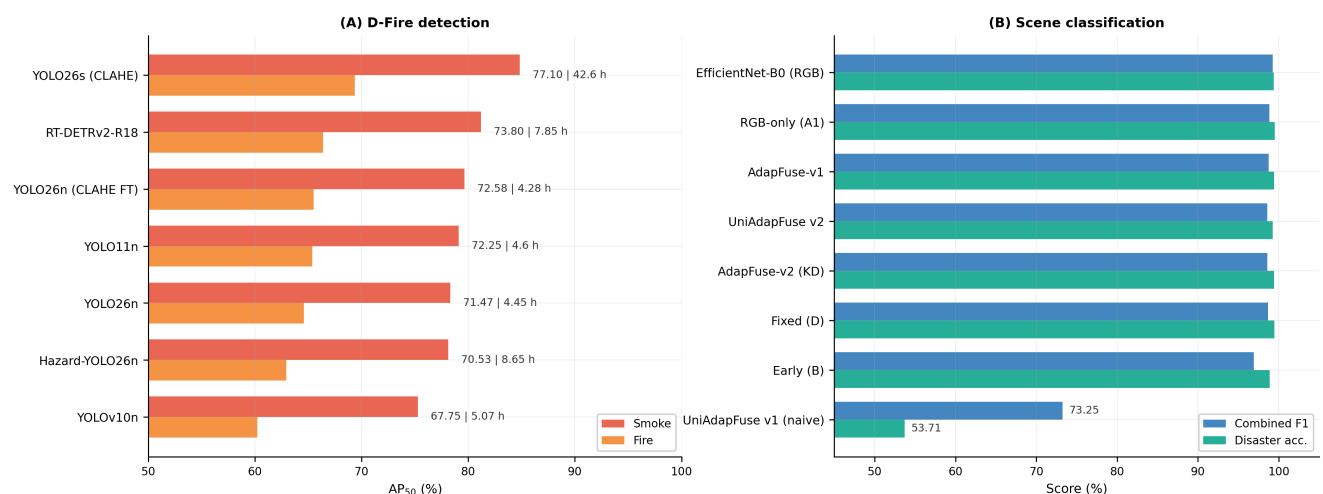


Figure 3 (thesis Fig. 5.2). (A) D-Fire detectors: class-wise AP with mAP₅₀ and summed training time. (B) Scene classification including the joint prototype: naive joint training collapses to 53.71% disaster accuracy; the revised recipe restores 99.26%.

7.2 The Hazard Prior Gate is a negative result

Same 40-epoch recipe, both runs complete: HPG reaches **70.53%** mAP₅₀ against **71.47%** for the baseline and needs **8.65 h vs 4.45 h** (≈ 770 s vs 400 s per epoch). Per-epoch validation tracks the baseline (0.703 vs 0.702 at epoch 31; 0.709 vs 0.707 at epoch 40), so there is neither faster convergence nor higher accuracy. An earlier "2.27× faster" figure came from a summary file that recorded only the last of three resumed segments (2.57 + 4.13 + 1.94 h).

7.3 Joint model: classification recovered, detection not

Model	Dis. F1	Dis. Acc	Vic. F1	Comb. F1	Det. mAP ₅₀
AdapFuse-v1 (classification only)	99.13	99.44	98.36	98.74	n/a
UniAdapFuse v1 (naive joint)	56.69	53.71	89.80	73.25	n/a
UniAdapFuse v2 (curriculum + detach + KD)	99.20	99.26	97.94	98.57	0.0

The naive run's log shows every batch skipped (non-finite loss) in 8 of 17 epochs; in v2's final logged epoch every train, validation and test batch was non-finite. So the collapse and the zero mAP₅₀ are at least partly numerical instability, not only negative transfer. Because v2 changed several things at once and distills from the very model it is compared with, the recovery cannot be credited to stop-gradient alone.

7.4 All 26 classification models (row-level test, n = 10,951)

#	Model	Type	Dis. F1	Vic. F1	Comb. F1
1	AdapFuse-v1 + EfficientNet-B0 (RGB)	Backbone	99.23	99.24	99.24
2	AdapFuse-v1 + MobileViT-XXS (RGB)	Backbone	98.81	99.16	98.99
3	A1 RGB-only	Unimodal	99.24	98.46	98.85
4	AdapFuse-v1 + EfficientNet-B0 (thermal)	Backbone	99.24	98.37	98.80
5	Ablation: no RUE	Ablation	99.38	98.12	98.75
6	AdapFuse-v1 (E)	Proposed	99.13	98.36	98.74
7	Ablation: no nuisance head	Ablation	99.23	98.21	98.72
8	D Intermediate fixed fusion	Fusion	99.11	98.28	98.69
9	C Late fusion	Fusion	99.16	98.17	98.67
10	AdapFuse-v1 + GDBlock audio	Backbone	98.88	98.38	98.63
11	F AdapFuse-v2 (KD student)	KD	99.17	98.05	98.61
12	UniAdapFuse v2 (joint)	Joint	99.20	97.94	98.57
13	RGB + pseudo-thermal subset	Two-sensor	98.70	98.30	98.50
14	RGB + audio subset	Two-sensor	98.96	98.02	98.49
15	AdapFuse-v1 + domain prompts	Module	98.85	98.13	98.49
16	AdapFuse-v2 + CNN6-style audio	Backbone	98.84	98.13	98.48
17	AdapFuse-v1 + spatial attention	Module	98.74	98.14	98.44
18	Ablation: no quality tokens	Ablation	98.81	98.00	98.40
19	Ablation: no cross-attention	Ablation	98.94	97.85	98.39
20	AdapFuse-v1 + CNN6-style audio	Backbone	98.58	97.92	98.25
21	Ablation: no GRU	Ablation	98.63	97.65	98.14
22	Pseudo-thermal + audio subset	Two-sensor	98.80	97.30	98.05
23	A2 Pseudo-thermal-only	Unimodal	98.02	96.84	97.43
24	B Early fusion	Fusion	98.32	95.54	96.93
25	UniAdapFuse v1 (naive joint)	Joint	56.69	89.80	73.25
26	A3 Audio-only	Unimodal	38.20	43.65	40.93

All F1 values are macro-F1 in %. Single seed per row. Rows 3 and 6 are highlighted: the RGB-only baseline sits above the proposed fused model.

7.5 Does fusion help? Matched-backbone comparison

Backbone	RGB-only	Pseudo-thermal-only	Fused (RGB stream)	Fused (thermal stream)
MobileNetV3-Small	98.85	–	98.74	–
MobileViT-XXS	98.99	98.46	98.99	–
EfficientNet-B0	98.88	98.60	99.24	98.80
ResNet-18	–	97.43	–	98.74

With the backbone held fixed, fusion adds 0.36, 0.00 and 0.20 points, and the unimodal runs had only a 12-epoch budget versus 50 for the fused models. None of this shows a fusion benefit: the extra inputs carry little beyond the RGB frame. For pseudo-thermal, EfficientNet-B0 beats ResNet-18 by 1.17 points at 21% of its MACs.

Audio-only. A3 reaches 71.69% disaster accuracy but 38.2% macro-F1: right on almost every row with audio, "fire" on every silent row (expected 71.73%), and "no person" everywhere (77.46% = share of no-person rows). The CNN6-style and GDBlock audio-only runs produced identical metrics to every digit, a shared degenerate solution.

7.6 Public benchmarks under published protocols

AIDER (Lee et al. 4:1:2 counts: 3,207 / 753 / 2,473 test, five classes, 71% normal):

Model	Source	Params (M)	F1 (%)	Acc. (%)
EfficientNet (scratch)	Lee et al. 2023	3.50	80.0	–
EmergencyNet	Lee et al. 2023	0.09	83.1	–
WATT-EffNet-3-6	Lee et al. 2023	0.69	88.5	–
RGB-only EfficientNet-B0	Ours (ImageNet init.)	4.67	94.00 [97.00 weighted]	97.01
RGB-only MobileNetV3-Small	Ours	1.22	93.02 [96.48]	96.48
AdapFuse-v1 (RGB + pseudo-thermal)	Ours	12.94	91.86 [95.79]	95.79

We exceed the published results under this protocol, but Lee et al. trained from scratch, so the margin shows ImageNet transfer works, not that our design is better. TakuNet and GlimmerNet (93.8–94.2% weighted F1, 0.03–0.04M params) use a different split and are 50–150× smaller. Weak classes: collapsed building (F1 86.0–88.7) and traffic incident (85.7–89.4; recall 86.6).

FLAME (official cross-camera protocol: train Matrice 200 / Zenmuse X4S, test 8,617 Phantom 3 frames):

Model	Source	Val. F1	Test acc.	Test F1	Fire rec.	No-fire rec.
Xception	Shamsoshoara 2021	–	76.23	–	–	–
EfficientNet-B5 + DenseNet-201	Ghali 2022	–	85.12	84.77	–	–
MV2-CBAM (30-run mean / 5-model ens.)	Deng 2025	–	85.82 / 88.05	–	–	–
FireXplainNet	Khan and Park 2024	–	87.32	86.50	88.87	88.01
RGB-only EfficientNet-B0	Ours	99.92	83.72	82.85	89.10	75.80
AdapFuse-v1	Ours	99.96	75.53	72.93	89.33	55.14
RGB-only MobileNetV3-Small	Ours	99.86	69.41	62.76	93.65	33.62

Above the original baselines, 1.4–4.3 points below three recent methods. Fire is found as reliably as the best model; the gap is clean frames called fire on the new camera (no-fire recall 75.8 vs 88.0).

D-Fire (published values as reported in each paper):

Source	Model	mAP ₅₀	AP _{smoke}	AP _{fire}
de Venâncio 2022	Pruned YOLOv4	73.98	–	–
de Venâncio 2023	YOLOv4 / YOLOv5l	76.56 / 79.46	83.48 / 86.38	69.94 / 72.84
Gonçalves 2024	YOLOv8n / YOLOv8s / YOLOv7x	77.80 / 78.70 / 80.40	84.0 / 84.9 / 85.5	71.5 / 72.5 / 75.4

Source	Model	mAP ₅₀	AP _{smoke}	AP _{fire}
Ruangsang 2026	YOLOv8m (3 seeds)	79.04 ± 0.21	–	–
Ours	YOLO26s (CLAHE)	77.10	84.85	69.36

Bottom line on state of the art: exceeds published AIDER results under Lee et al.'s protocol; competitive with, but below, recent FLAME and D-Fire methods. Never claim SOTA.

7.7 Generalisation reality check

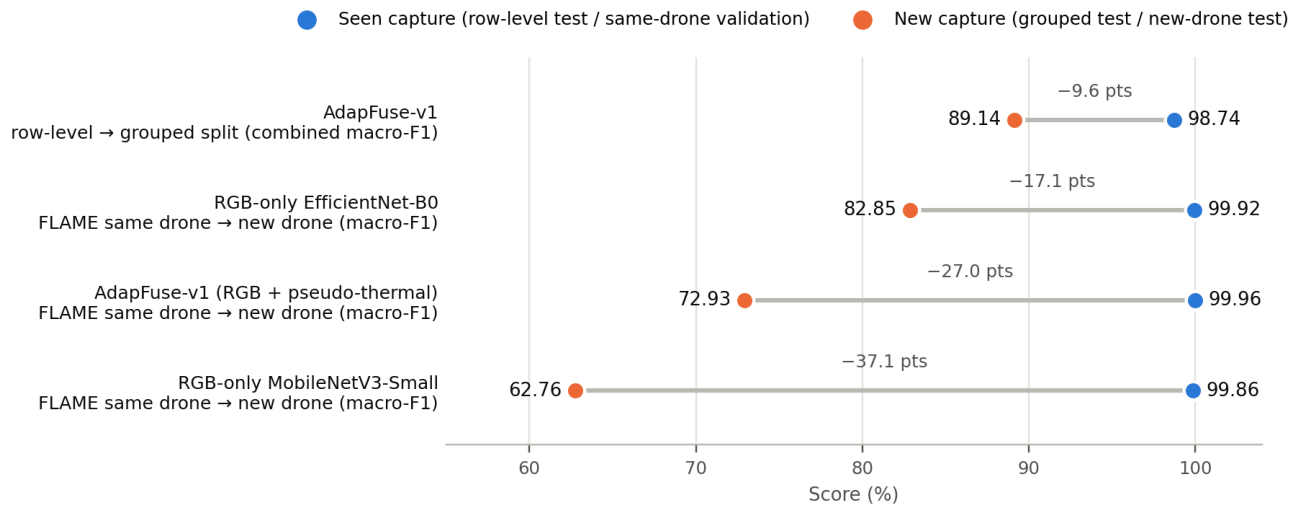


Figure 4 (new for this guide). Scores on captures the model has seen (blue) against new captures (orange). Every model loses 9.6–37.1 points once test frames come from a different capture group or a different drone. Sources: thesis Tables 5.11 (FLAME) and 5.14 (grouped split).

AdapFuse-v1	Row-level (n = 10,951)	Grouped (n = 11,097)	Δ
Disaster macro-F1	99.13	84.85	−14.28
Victim macro-F1	98.36	93.43	−4.93
Combined macro-F1	98.74	89.14	−9.60
Disaster accuracy	99.44	89.05	−10.39
Test focal loss	0.0020	0.1462	×73

Caveats. The grouped run also used batch size 2 and no class balancing; its weakest classes are the rare ones (collapse/flood 0.824, other 0.765). Only seed 42 gave a usable result (seed 41 picked a checkpoint inside a 16-epoch window of degenerate 1.0000 validation; seed 43 stopped at epoch 12). The −9.60 mixes leakage with recipe effects.

Source-label confounding (grouped test set)

Source	n	Disaster classes	Victim classes	Dis. acc.	Vic. acc.
AIDER	965	0, 1, 2, 3	0 only	84.15	57.20
C2A	1,535	1, 2, 3	1 only	81.69	95.90
FLAME	7,244	0, 1	0 only	89.61	99.94
ESC-50	400	0, 1	0 only	97.50	100.0
SARD	863	0 only	1 only	99.77	99.30
FireNet	90	1 only	0 only	82.22	60.00

Victim presence is fully determined by the source; SARD is all clean, FireNet all fire. A model can score well by recognising the dataset. AIDER's 57.2% victim accuracy (people predicted in person-free disaster scenes) fits the model linking disaster appearance with C2A. The low per-source macro-F1 written by the evaluation script (0.30–0.56) is an artefact of scoring absent classes as zero.

7.8 Reliability gates and corruption stress tests

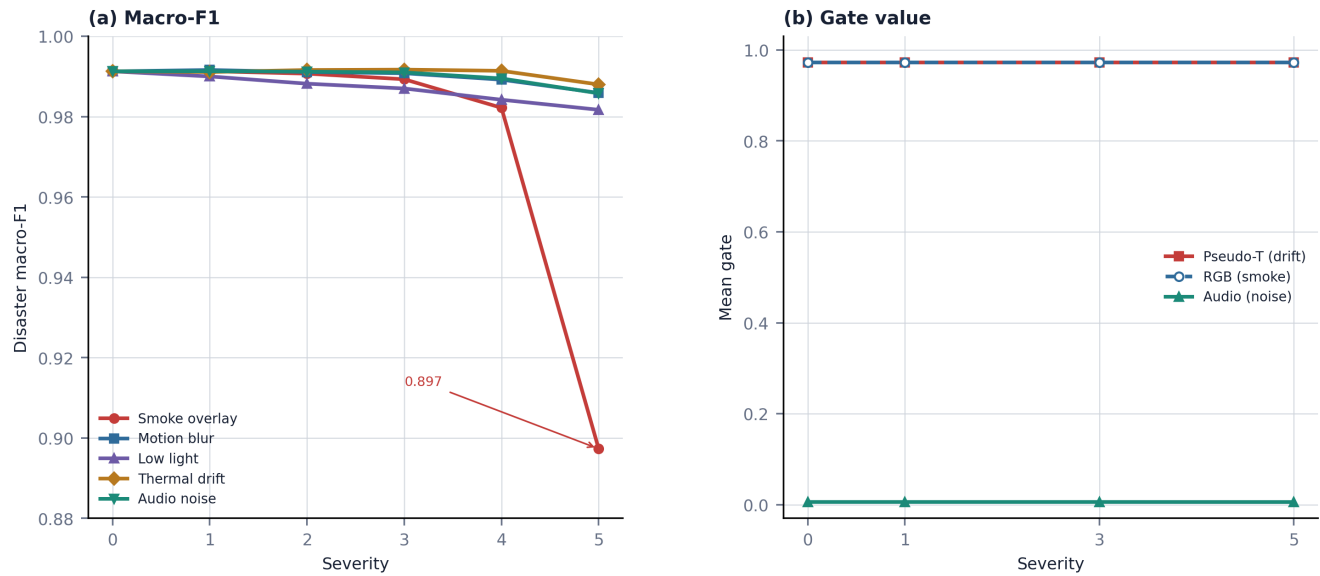


Figure 5 (thesis Fig. 5.3). (A) Disaster macro-F1 under five synthetic corruptions; severe smoke reduces it to 0.897. (B) Mean gates stay flat across severity (0.972 = share of rows with RGB, i.e. ≈ 1.0 on every row with input).

Corruption (severity 5)	Disaster macro-F1	Gate behaviour
Clean	99.13	RGB ≈ 1.0 , pseudo-thermal ≈ 1.0 , audio ≈ 0.011 (rows with input)
Synthetic smoke overlay	89.73	RGB gate unchanged
Low light	98.17	unchanged
Audio noise	98.58	audio gate already ≈ 0
Motion blur	98.59	unchanged
Pseudo-thermal drift	98.80	unchanged

The grouped retraining learned different gates: audio 1.0 on all 5,982 rows with audio (consistent with using the label shortcut) and pseudo-thermal bimodal, off on 43.5% of rows, mostly FLAME. Gates are run-dependent on/off switches; robustness comes from redundancy, not dynamic reweighting.

7.9 Component ablations

Variant	Component removed	Comb. F1	Δ (pp)	Test loss $\times$
AdapFuse-v1 (base)	–	98.74	–	1.0
no GRU	single-step GRU	98.14	–0.60	5.1
no attention	4-head attention $\rightarrow$ mean	98.39	–0.35	3.8
no quality tokens	16-d quality heads	98.40	–0.34	5.4
no nuisance	unsupervised nuisance head	98.72	–0.02	4.6
no RUE	reliability gating	98.75	+0.01	4.7

Single runs; the base model used the cycling lr schedule while all ablations used single cosine decay. The no-noisance run changes no training signal, so its 4.6 $\times$ loss change with -0.02 pp is a direct measure of run-to-run noise. Treat these as hypotheses.

7.10 Calibration, false positives and latency

Metric (AdapFuse-v1, row-level)	Value	Module (batch 1, RTX 5090)	ms
Disaster precision / recall	98.70 / 99.57	RGB backbone (MobileNetV3-S)	6.37
False-positive rate	0.73%	Pseudo-thermal backbone (ResNet-18)	5.17
Expected calibration error	0.0337	Audio backbone	0.47
Test focal loss	0.0020 (lowest)	RUE	0.85
Parameters	12.94M (v2: 12.30M)	Projectors + quality + attention + GRU + nuisance	1.40
Throughput	69.9 FPS	End-to-end	14.32

Backbones take 83.8% of the time. At batch 1 on a large GPU latency is kernel-launch-bound, so these timings do not transfer to an embedded device; no embedded, INT8 or power measurement was made.

7.11 Requirements traceability (Chapter 3 targets)

Requirement	Target	Evidence	Status
Wildfire detection	≥95%	99.44% disaster classification accuracy; 77.10% mAP ₅₀ detection	Met for classification only
Victim detection	≥90%	98.36% victim-presence macro-F1; no person boxes evaluated	Met for classification only
False-positive rate	≤8%	0.73% row-level scene FPR	Met on row-level split
Graceful degradation	≥85% with one modality missing	Two-sensor subsets 98.05–98.50%	Partially met (pseudo-thermal not independent)
Latency	≤500 ms at ≤15 W	14.32 ms on RTX 5090	Not validated
INT8 quantisation	All encoders	Not implemented	Not met
Synchronised dataset	≥6,000 tri-modal sequences	Pseudo-thermal + label-matched audio	Not met

8. The demo application

Built at the supervisor's request in `code_p2/adaptfuse_uav/demo_app`: upload one video, RGB or thermal, with its audio track. Two web front-ends share one inference engine and a pytest suite.

- **Live app** (FastAPI + WebSocket, port 8000): the browser plays the video with sound and sends the current frame with its timestamp; the server cuts the matching 2 s audio window and returns boxes and labels drawn over playback. About 12.5 analyses/s on the development GTX 960.
- **Batch app** (Gradio, port 7860): returns an annotated MP4, a timeline plot, a summary and per-frame CSV/JSON.
- **Stability**: EMA smoothing of class probabilities ($\tau = 0.5$ s), boxes drawn only if seen in 2 of the last 3 frames, and a "models disagree" badge when the detector and scene classifier contradict each other.

Output	Model	Provenance
Disaster class, victim yes/no, RUE gates	Ensemble of 2 grouped-split AdapFuse-v1 seeds (41, 42) + flip TTA + val-tuned class bias	Trained in our pipeline
Zero-shot hazard events and zones	CLIP ViT-B/16 vision tower + prompt-ensemble text embeddings; fused with AdapFuse (weight 0.35 RGB / 0.40 thermal)	Pretrained; bias, weight and temperatures tuned on grouped val
Fire / smoke boxes	YOLO26s fine-tuned on CLAH E-Fire (77.1% mAP ₅₀); class-name guard {0: smoke, 1: fire}	Trained in our pipeline
Person boxes	YOLO26n, COCO person class	COCO-pretrained, not fine-tuned
Sound tags	PANNs Cnn14-16k (AudioSet mAP 0.438; Cnn6 optional)	AudioSet-pretrained, not fine-tuned
Audio-only label	AudioCNN baseline (A3)	Trained in our pipeline

8.1 Demo accuracy (grouped split, seed-42 metadata)

Scene classifier stage	RGB disaster macro-F1	Thermal disaster macro-F1
Single seed (41 / 42)	0.869 / 0.890	0.348 / 0.264
Ensemble + flip TTA + bias (classifier in app)	0.919	0.355 (collapse/other never predicted)
Zero-shot CLIP + val bias (test subset)	0.895	0.825
Fused, current app (test subset)	0.967	0.852
Real-world clips (9 clips, 72 frames): AdapFuse → fused	0.119 → 0.475 macro-F1 (frame acc. 0.208 → 0.944)	0.115 → 0.422

The full-test classifier numbers use all 10,697 rows; the zero-shot and fused numbers use a stratified subset (≤ 150 images per dataset and class, $n = 1,597$). The real-clip set has only fire, flood and collapse frames, so its 4-class macro-F1 is capped; it is a sanity check, not a benchmark. The trained classifier's collapse on real clips (0.119) is the same generalisation gap the thesis measures.

9. Contributions, limitations and future work

9.1 What we can defend as contributions

- An implemented, open, tested tri-modal architecture that runs with any sensor subset, plus a joint classification + detection prototype with a documented failure mode.
- A physics-prior detection module evaluated honestly to a negative result, with corrected timing.
- An audit of reliability and nuisance mechanisms: the gates saturate, do not track corruption, and differ between runs; nuisance is unsupervised.
- Reproducible evidence that assembled public corpora inflate multimodal scores: -9.60 pp under grouped splitting, $17\text{--}37$ pp drops across cameras on FLAME, source-determined victim labels, and label-leaking audio.
- Public-protocol results: above published AIDER results under Lee et al.'s protocol; below recent FLAME and D-Fire methods.
- A working demo with explicit model provenance.

9.2 Main limitations

- No synchronised or independent sensors: pseudo-thermal from RGB; label-matched, label-leaking audio.
- Single seeds; differences under ~ 1 point are within noise; the lr schedule differs across some compared runs.
- RUE without degradation targets; GRU on single frames; nuisance head without labels.
- Joint detector $\text{mAP}_{50} = 0.0$ with non-finite losses; no person-box benchmark.
- No embedded latency, power or INT8 results.

9.3 Future work, in priority order

- Synchronised data: FLAME 2 RGB-IR pairs, DroneAudioset or DREGON audio; rebuild audio pairing so the clip class does not encode the label.
- Controlled grouped-split rerun with the row-level recipe, three or more seeds, leave-one-source-out evaluation and sources containing both values of each label.
- Supervise the RUE with degradation labels and pre-backbone quality signals (dark-channel prior, blur, SNR); test gate response.
- Diagnose the joint detector (loss weighting, head lr, target and evaluation path, non-finite losses).
- Fine-tune person detection on SARD; run the GRU on multi-frame sequences.
- Profile AdapFuse-v2 on embedded hardware with INT8 quantisation; zero-shot tests on HIT-UAV and FLAME 2.

Part B · Defense questions and model answers

Answer in this order: the direct answer first, then the number, then the caveat. Panels reward a candidate who states a limitation before being pushed. Lines marked **Avoid** are phrases that the thesis evidence does not support. Panels also ask each member what they personally did, so agree on who answers which topic (data, architecture, detection, demo) before the day.

B1. Motivation, scope and contribution

Q1. In one or two sentences, what is your thesis about?

We built and evaluated UniAdapFuse, a framework that fuses RGB, pseudo-thermal and audio inputs to classify the disaster type and victim presence from UAV data, gating each modality with a learned reliability estimator so it runs with any subset of sensors. We also benchmarked smoke/fire detectors on D-Fire, tested a physics-guided Hazard Prior Gate, and audited what public data can show about fusion.

Q2. What is new compared with existing fusion work such as RGB-thermal wildfire models?

Three design elements and one evaluation contribution: (1) an RUE that gates per-modality tokens before cross-modal attention, with a presence mask so one model handles any sensor subset; (2) the Hazard Prior Gate, a physics prior in front of a YOLO detector; (3) a joint classification + detection graph with a phased curriculum, stop-gradient and distillation; (4) a systematic audit (grouped split, per-source decomposition, audio-leak analysis).

We are explicit that (1)–(3) are implemented but not shown to improve accuracy; (4) is our strongest result.

Q3. Your fused model does not beat RGB-only. Why is this a thesis and not a failed project?

Because the negative result is measured carefully and explained. The pseudo-thermal stream is computed from RGB and the audio is label-matched, so the fused inputs carry almost nothing beyond the RGB frame. Under those conditions no fusion method should beat RGB-only, and it doesn't (98.74% vs 98.85%; matched backbones differ by 0.00–0.36 points).

We also show that near-ceiling scores fall 9.60 points under a grouped split and that the source determines the victim label. That defines what a credible fusion benchmark needs: synchronised sensors, degradation labels and source-decorrelated splits.

Avoid: "Our fusion improves accuracy."

Q4. Why UAVs, and why three modalities?

UAVs deploy quickly with sub-metre resolution, while satellites revisit every 5–16 days and ground sensors only cover points. The three sensors fail differently: RGB under smoke, dark and glare; thermal at low resolution and with sun-heated false alarms; audio under rotor noise. Reliability-weighted fusion is meant to cover each failure with another sensor. That motivation stands; our data just cannot test it yet.

Q5. Why is it called UniAdapFuse if the unified detector does not work?

The name refers to the architecture, one graph for classification and detection. In that graph classification recovers (98.57% combined F1) but detection records $\text{mAP}_{50} = 0.0$, so we present it as a prototype with a documented failure mode. All working detection results come from separately trained detectors.

B2. Data

Q6. Describe your dataset.

72,997 rows from six public sources: FLAME 47,992 (65.7%), C2A 10,215, AIDER 6,433, SARD 5,755, ESC-50 2,000 audio clips and FireNet 602. Each row has a 4-class disaster label (clean, fire/smoke, collapse/flood, other), a binary victim-presence label and presence flags for three modalities. Row-level split 51,097 / 10,949 / 10,951, stratified by disaster label.

Q7. Is your thermal data real thermal imagery?

No. All 70,997 thermal maps in our metadata are pseudo-thermal, computed from the RGB frame, not radiometric LWIR. So RGB + thermal fusion here combines two views of one sensor: it cannot see heat sources invisible to RGB and it degrades with RGB. We say "pseudo-thermal" throughout; independent LWIR (for example FLAME 2 RGB-IR pairs) is future work.

Avoid: "thermal sensing", "night-time fallback", "sees through smoke".

Q8. How was audio paired with images, and what is the label leak?

ESC-50 clips were assigned to image rows by label, not recorded together; 2,000 clips are reused across 39,003 rows. Every fire/smoke row with audio gets one of the 40 "crackling fire" clips, every clean row one of the other 1,960, and collapse/flood and other rows have none. So on rows with audio the clip class identifies fire versus clean perfectly, a shortcut any model can learn. It holds in both splits.

Q9. If audio leaks the label, why does audio-only score only 38.2% macro-F1?

Because 47.2% of test rows have no audio and collapse and other never have audio. A3's 71.69% disaster accuracy equals being right on almost every row with audio and predicting fire on every silent row (expected 71.73%). Its 77.46% victim accuracy equals the share of no-person rows. The low macro-F1 reflects missing inputs and absent classes, not uninformative audio.

Avoid: "audio is uninformative".

Q10. Why balance classes, and doesn't 48% duplicated training data cause overfitting?

The raw training split is imbalanced (fire 46.5%, other 4.0%). We oversampled to class balance and 50/50 victim balance, then took a source-stratified half: 49,276 rows, 12,319 per class, 48.1% duplicates. Duplicates get stronger augmentation, the test set is never resampled, and we report macro-F1. Duplication can still encourage memorising minority images, one reason we also ran the grouped split.

Q11. What is the difference between the row-level and grouped split, and why does it matter?

The row-level split assigns frames at random, so near-identical neighbouring frames of one FLAME video can be in both train and test. The grouped split keeps whole capture groups on one side (9,267 / 2,003 / 2,002 groups) and partitions ESC-50 by fold. AdapFuse-v1 drops from 98.74% to 89.14% combined macro-F1 (−9.60; disaster −14.28, victim −4.93), so row-level numbers are within-collection estimates.

Q12. FLAME has no video IDs. How did you build its groups?

We blocked frames into contiguous runs of 250 as an adjacent-frame proxy; C2A augmentations and SARD variants are grouped by original image. It is a proxy: frames across block boundaries or from repeated passes can still look alike, so even the grouped score may be optimistic.

Q13. Is the 9.60-point drop purely leakage?

No. The grouped run also used batch size 2 and no class balancing, and its weakest classes are the rare ones (collapse/flood 0.824, other 0.765), which imbalance alone would produce. Only one of three seeds was usable. A controlled rerun with the row-level recipe is needed to isolate leakage.

Q14. What is source-label confounding?

In the grouped test set victim presence is fully decided by the source: every C2A and SARD row has a person, every AIDER, FLAME, ESC-50 and FireNet row has none; SARD is all clean and FireNet all fire. A model can score well by recognising the dataset (resolution, compression, colour, altitude). Evidence: AIDER victim accuracy is only 57.2%, with people predicted in person-free disaster scenes. Fixes: sources with both label values, source-balanced sampling, leave-one-source-out evaluation.

Q15. C2A images are synthetic. Does that matter?

Yes. C2A pastes human figures onto UAV disaster scenes, so its person-visible label marks pasted figures and can be learned from compositing artefacts. With the source confound, our victim-presence scores are not evidence of person detection.

Q16. Why are the per-source F1 scores so low (0.30-0.56) in the grouped table?

A metric artefact: the script averages over all classes with `zero_division = 0`, so a class absent from a source scores zero and a perfect single-class source caps at 0.50 victim macro-F1. Accuracy (disaster 0.82-0.998) and present-class F1 show within-source classification is reasonable.

B3. Architecture

Q17. Walk us through AdapFuse-v1.

Three encoders: MobileNetV3-Small for RGB, 1-channel ResNet-18 for pseudo-thermal, a 3-layer 2D-CNN for 64×63 log-mel audio. Each gives a 192-d token and a 16-d quality token. The RUE, a 2-layer MLP over the quality tokens plus 6 feature statistics, outputs sigmoid gates; each token is scaled by its gate (zero if missing). 4-head attention mixes the tokens, mean pooling gives one vector, a single-step GRU refines it, and heads predict disaster (4) and victim (2); a 5-class nuisance head is defined but unsupervised. 12.94M parameters, 14.32 ms on the workstation GPU.

Q18. What did the RUE actually learn?

Static on/off switches. In the row-level model RGB and pseudo-thermal gates are ≈ 1.0 on every row with input and audio ≈ 0.011 , unchanged across five corruptions and five severities. The grouped retraining learned the opposite for audio (1.0 on all 5,982 rows) and switched pseudo-thermal off on 43.5% of rows, mostly FLAME, which looks like a response to dataset identity. So the RUE is a run-dependent weighting, not a demonstrated reliability estimator.

Avoid: "the RUE detects sensor degradation".

Q19. Why don't the gates react to corruption?

Nothing in training ties a gate to sensor quality: degradation targets are None, and `corruption_prob = 0.3` is set without a `corruption_type`, so no corruption is applied while training. The only signals are task loss and a $-\text{Var}(r)$ regulariser, which pushes gates apart towards saturation. Dynamic gates need degradation labels or corruption-augmented training, pre-backbone quality signals, and a test against measured degradation.

Q20. Then on what basis do you report robustness under smoke?

Only what the sweep shows: disaster macro-F1 falls from 99.13% to 89.73% at severity-5 synthetic smoke and stays at or above 98.17% for blur, low light, pseudo-thermal drift and audio noise. Because the gates are flat, we attribute this to residual visual signal and redundancy with the RGB-derived map. There is no matched no-RUE corruption baseline, so we do not claim the RUE causes it.

Q21. Why use a GRU when every sample is one frame?

It was designed for temporal smoothing, but the loader gives independent frames, so it runs with sequence length 1 from a zero state and acts as a gated nonlinear layer. Removing it cost 0.60 points in one run, which is not evidence of temporal modelling. Multi-frame training is future work; the demo gets temporal stability from EMA smoothing and box persistence instead.

Q22. What does the NuisanceAwareHead do without labels?

Nothing supervised: nuisance_gt is None, so its loss is zero. It is an architectural provision. The no-nuisance ablation therefore changes no training signal and acts as a re-run, which is useful: it shows a 4.6× test-loss change with −0.02 pp F1, our measure of run-to-run noise. The 0.73% FPR cannot be credited to it.

Q23. Is attention over three tokens meaningful?

It is modality-level interaction between three pooled vectors, not spatial correspondence. Replacing it with mean pooling cost 0.35 points in one run, within noise. A 7×7 spatial-attention variant (98.44%) did not help. With pseudo-thermal derived from RGB, spatial RGB-thermal alignment cannot be tested anyway.

Q24. How did you choose the backbones?

Four criteria: public pretrained weights; a per-stream budget of about 1 GMAC and 5M parameters because three encoders run per frame; best ImageNet top-1 inside the budget; three design families. That selects MobileNetV3-Small (0.06 GMACs), EfficientNet-B0 (0.39 GMACs, 77.7% top-1) and MobileViT-XXS. We measured 13 candidates with one script. ResNet-18 is over budget and kept only as the thermal baseline; EfficientNet-B0 beats it on pseudo-thermal (98.60% vs 97.43%) at 21% of the MACs.

Q25. Why is early fusion (Design B) the weakest fusion design?

Partly topology: one ResNet-18 must process RGB, pseudo-thermal and a resized spectrogram as five stacked channels, losing modality-specific features.

But in the code Design B's ResNet-18 is built with weights=None, trained from scratch with a new 5-channel conv1, while all other designs use ImageNet weights. Its 96.93% therefore mixes topology with the lack of pretraining, and we cannot attribute the gap to early fusion alone.

Avoid: attributing the gap only to early fusion (and disclose the scratch initialisation in the thesis, fix list item 2).

Q26. Explain the "PANNs" / CNN6-style audio encoder.

It was meant to transfer PANNs CNN6 AudioSet weights, but our implementation uses 3×3 kernels and the checkpoint has 5×5, so no convolution weights load; only 8 of 20 tensors (batch norm) do. It is effectively trained from scratch, which is why the thesis calls it CNN6-style. Alone, it collapses to the same degenerate predictions as GDBlock. Pretrained audio transfer was not tested.

Avoid: "AudioSet-pretrained PANNs transfer".

B4. Training

Q27. What were the training settings?

AdamW (lr 10^{-3} , weight decay 10^{-4}), 5-epoch warm-up then cosine to 10^{-5} , batch 16 (8 for EfficientNet-B0), up to 50 epochs with patience 15, fp16 AMP, focal loss (α 0.25, γ 2) for disaster and victim plus a reliability-variance term, and 20% random single-modality dropout. Trained on an RTX 5090 (32 GB); about 390 GPU-hours in the logs.

Q28. Were all compared runs trained identically?

Not entirely, and we disclose it. The logged lr cycles every ~8 epochs for C, D, E, F, domain prompts and naive UniAdapFuse, while A1 and all five ablations follow a single cosine decay. UniAdapFuse v2 used its own recipe, the unimodal backbone variants had a 12-epoch budget, and the grouped run used batch 2 without balancing.

Q29. Why focal loss?

The test set keeps the natural imbalance (other hazard 4.0%). Focal loss down-weights easy majority examples so rare classes still drive the gradient; together with balanced sampling and macro-F1 reporting it keeps rare classes from being ignored.

Q30. How many seeds? Are your differences significant?

One seed per configuration on the row-level split, with no confidence intervals. Differences under about one point (fusion vs RGB-only 0.11; ablations 0.34–0.60) are within run-to-run variation, as the no-nuisance re-run shows. We call them descriptive associations, not causal effects.

Avoid: "significantly better".

B5. Classification results

Q31. Why macro-F1 and Combined F1 rather than accuracy?

The test set is imbalanced: collapse (8.5%) and other (4.0%) are only 12.5% of rows, so accuracy can stay above 98% while a model fails on both. Macro-F1 weights every class equally. Combined F1 is the plain mean of disaster macro-F1 and victim macro-F1, so both tasks count equally.

Q32. What is your best classification result?

AdapFuse-v1 with an EfficientNet-B0 RGB encoder: 99.24% combined macro-F1 on the row-level split (disaster 99.23%, victim 99.24%). Base AdapFuse-v1: 98.74% (disaster 99.13%, victim 98.36%, accuracy 99.44%). Always add: row-level, single seed, within-collection.

Q33. Does fusion improve accuracy?

Not on this data. RGB-only (98.85%) is 0.11 above AdapFuse-v1 (98.74%), and the no-RUE ablation (98.75%) matches it. With the backbone fixed, fusion adds 0.36, 0.00 and 0.20 points, with shorter budgets for the unimodal runs. On AIDER and FLAME AdapFuse-v1 is below RGB-only EfficientNet-B0. That is expected when the extra inputs are RGB-derived or label-matched.

Q34. What do the ablations show?

Removing the GRU, attention or quality tokens lowers combined F1 by 0.60, 0.35 and 0.34; removing the RUE gives +0.01; removing the unsupervised nuisance head −0.02. All single runs with a different lr schedule from the base, so they are hypotheses for controlled follow-up, not component effects.

Q35. What does knowledge distillation (AdapFuse-v2) give you?

Halving the projection width (192 → 96) with logit KD ($T = 4$) and feature KD ($\lambda = 0.5$) costs under 0.2 points (98.61%) but removes only 4.9% of parameters (12.94M → 12.30M), because the backbones dominate. Real compression needs smaller or quantised backbones; we did not measure the student on embedded hardware.

Q36. What do the ECE of 0.0337 and the 0.73% FPR mean?

On the row-level test set AdapFuse-v1's confidence matches its accuracy to within 3.4% on average, and 0.73% of non-hazard scenes are flagged. Both are row-level statistics, not field false-alarm rates, and the FPR cannot be credited to the unsupervised nuisance head.

Avoid: "0.73% false-alarm rate in the field".

B6. Detection and the Hazard Prior Gate

Q37. Why evaluate detection on D-Fire rather than your own corpus?

Our corpus has no smoke/fire boxes; its only boxes are person boxes in C2A and SARD. D-Fire has 21,527 images with smoke and fire boxes. The detectors are trained independently, not as UniAdapFuse heads, and no person detector was trained.

Q38. What are your detection results?

Best: YOLO26s with CLAHE at 640 px, SGD, 100 epochs, 77.10% mAP_{50} (smoke 84.85, fire 69.36; $mAP_{50:95}$ 44.80). Then RT-DETRv2-R18 73.80, YOLO26n CLAHE fine-tune 72.58, YOLO11n 72.25, YOLO26n 71.47, Hazard-YOLO26n 70.53 and YOLOv10n 67.75. Smoke beats fire by 13.7–15.5 points in every model.

Q39. Why is fire AP so much lower than smoke AP?

Flames are often small, partly occluded and confusable with sunsets, lights and reflections, while smoke plumes are large, high-contrast regions; small objects are hardest at 512–640 px. Our fire gap to the best published model (YOLOv7x 75.4) is 6 points, larger than the smoke gap.

Q40. What is the Hazard Prior Gate and why didn't it help?

HPG computes $\text{fire} = \text{ReLU}(R - (G+B)/2)$, smoke achromaticity $= 1 - (\max - \min)/\max$ and a Sobel edge map, pools them to quarter resolution, encodes them with a small CNN and adds a bounded residual to the input: $x + s \cdot g \cdot \Delta x$ with $s \leq 0.25$. Same 40-epoch recipe: 70.53% vs 71.47% mAP_{50} , 8.65 h vs 4.45 h.

A detector trained on 15k+ labelled images already learns colour and edge cues, so a hand-crafted input correction adds little, and computing the cues at full resolution adds cost. It might help with little data or with real thermal input; that is untested.

Q41. An earlier version said HPG trained 2.27× faster. What happened?

The HPG run was interrupted and resumed twice, and its summary file recorded only the last segment (1.96 h). The per-epoch results.csv shows three segments, $2.57 + 4.13 + 1.94 = 8.65$ h. With the correct total HPG is about 1.9× slower, and its validation curve tracks the baseline. We corrected the thesis and report HPG as a negative result.

Avoid: any speed-up claim for HPG.

Q42. Does CLAHE help?

We cannot isolate it. The YOLO26s CLAHE run also changes model size, image size (640 vs 512), optimiser, epochs (100 vs 40) and batch size. The YOLO26n CLAHE fine-tune adds 1.11 points (71.47 → 72.58) but also gets 20 extra epochs at 640 px. A same-recipe CLAHE vs no-CLAHE run is needed.

Q43. How do your detectors compare with published D-Fire results?

Below recent work: YOLOv7x 80.40, YOLOv5l 79.46, YOLOv8m 79.04 ± 0.21 and YOLOv8n 77.80, against our 77.10; we beat only YOLOv4 (76.56 / 73.98). The gap is mostly fire AP and our short nano recipes (512 px, 40 epochs, about six points under the published YOLOv8n). Our D-Fire numbers are internal baselines for the HPG comparison.

Published values are copied from the papers; we did not re-run them. Our test set is the official 4,306-image split (fix list item 1).

Avoid: "we re-evaluated all published methods on our split".

B7. The joint model

Q44. What happened in naive joint training?

UniAdapFuse v1, trained end-to-end for classification and detection, fell to 53.71% disaster accuracy against 99.44% standalone. Our hypothesis was negative transfer from detection gradients, but the log shows every batch skipped in 8 of 17 epochs because the loss was NaN or Inf, so numerical instability is at least part of the cause.

Q45. How did v2 recover classification, and can you credit stop-gradient?

v2 combined a 15-epoch classification warm-up, `.detach()` on pyramid features entering the detection neck, differentiated learning rates, distillation from AdapFuse-v1, $\text{lr } 5 \times 10^{-4}$, 80 epochs and loss reweighting. Classification reached 99.26% accuracy and 98.57% combined F1. Since all changes came together and the teacher is the model we compare against, we cannot credit stop-gradient alone.

Avoid: "stop-gradient solved negative transfer".

Q46. Why is the joint detector's mAP_{50} exactly zero?

`.detach()` only blocks detection gradients from the backbone; the neck and head still train, so isolation alone doesn't explain it. In v2's final logged epoch every train, validation and test batch was non-finite. Candidates: unstable detection loss, the small detection weight (0.2), or a bug in target assignment or evaluation. Diagnosing it is first on our list; until then we don't claim joint perception.

B8. State of the art and public benchmarks

Q47. Is your work state of the art?

No, and we don't claim it. Under Lee et al.'s AIDER protocol RGB-only EfficientNet-B0 reaches 94.00% macro-F1 against 88.5% for WATT-EffNet, but we use ImageNet pretraining and they trained from scratch, so it shows transfer works. On FLAME's cross-camera test we reach 83.72% accuracy, above Xception (76.23) and ResNet-50 (79.48) but below 85.12–88.05. On D-Fire we are 0.7–3.3 points below published detectors.

Avoid: "state of the art" or "outperforms existing methods".

Q48. Why do your FLAME scores fall from 99.9% to 62–84%?

FLAME trains on a Matrice 200 with a Zenmuse X4S and tests on a Phantom 3. Same-drone validation is 99.86–99.96% for every model; new-drone test F1 is 62.76–82.85%. Fire recall stays high (89–94%) but no-fire recall drops to 33.6–75.8% (FireXplainNet 88.0): on the new camera our models call clean scenes fire, the training majority (64%). Validation saturates at epoch 1, so it cannot guide model selection.

Q49. Which AIDER classes are weakest, and why?

Collapsed building (F1 86.0–88.7%) and traffic incident (85.7–89.4%). For traffic, recall (86.6%) is the limit: accidents are confused with normal scenes, consistent with training without resampling on a 71%-normal set.

B9. Deployment and efficiency

Q50. Is your system real-time on a UAV?

Not demonstrated. AdapFuse-v1 runs in 14.32 ms (69.9 FPS) at batch 1 on an RTX 5090 workstation, far above the ≤ 15 W target. We did not profile embedded hardware or implement INT8. At batch 1 on a large GPU latency is kernel-launch-bound, which is why MobileNetV3-Small (6.37 ms) is slower than ResNet-18 (5.17 ms) and the small RUE MLP costs 0.85 ms, so these timings don't transfer to an embedded device.

Avoid: "real-time on a UAV" or "edge-ready".

Q51. Where does the inference time go?

The three backbones take 83.8% (RGB 6.37, pseudo-thermal 5.17, audio 0.47 ms); RUE, attention, GRU and nuisance head add 1.63 ms. Cutting cost means smaller or shared backbones and quantisation, not removing the fusion modules.

B10. The demo application

Q52. What does the demo do?

Built at the supervisor's request: upload one video, RGB or thermal, with its audio, and see the scene class, victim presence, fire/smoke boxes, person boxes and sound tags. The live FastAPI + WebSocket app overlays results during playback (about 12.5 analyses/s on a GTX 960); the Gradio app returns an annotated MP4, a timeline, a summary and per-frame CSV/JSON. Both share one engine with a pytest suite.

Q53. Which demo models are yours and which are pretrained?

Trained in our pipeline: the scene classifier (two grouped-split AdapFuse-v1 seeds + flip TTA + val-tuned bias), the audio-only AudioCNN and the D-Fire YOLO26s CLAHE detector. Pretrained and not fine-tuned: YOLO26n COCO for people, PANNs Cnn14-16k for AudioSet sound tags and a CLIP ViT-B/16 zero-shot hazard model. The app labels provenance, and the thesis results don't depend on the pretrained parts.

Q54. Why add a CLIP zero-shot model?

On nine real-world clips (72 frames) the trained ensemble scored 0.119 coarse macro-F1: it barely generalises beyond its training sources, the same gap the thesis measures. Zero-shot CLIP with prompt ensembles and val-tuned bias reaches 0.895 on the grouped test subset, and fusing both (weight 0.35 on AdapFuse, temperatures fit on val) reaches 0.967, with 94.4% frame accuracy on the real clips. The real-clip set is tiny and lacks normal and other frames, so treat it as a sanity check.

Q55. How good is the demo in thermal mode?

Weak for the trained classifier: thermal + audio gives 0.355 disaster macro-F1 on the grouped test, never predicting collapse or other, because training used RGB-derived pseudo-thermal maps and the D-Fire detector is RGB-only. With the zero-shot fusion it reaches 0.852 on the test subset. The app warns the user in thermal mode.

Q56. Your demo once drew smoke as fire. What happened?

The first bundled detector stored class names {0: fire, 1: smoke} while D-Fire uses {0: smoke, 1: fire}. We replaced it with our best YOLO26s CLAHE checkpoint (mAP₅₀ re-verified at 0.7711) and added a guard that refuses any detector whose names are not {0: smoke, 1: fire}.

B11. Requirements, ethics and impact**Q57. Did you meet your Chapter 3 requirements?**

Partly. On the row-level split, classification meets $\geq 95\%$ wildfire (99.44% accuracy), $\geq 90\%$ victim (98.36% F1) and $\text{FPR} \leq 8\%$ (0.73%). Graceful degradation is partly met (two-sensor subsets 98.05–98.50%, but pseudo-thermal is not independent). Latency at ≤ 15 W is not validated; INT8 and the synchronised dataset are not met. Classification accuracy is not detection accuracy: detection is 77.10% mAP₅₀ with no person-box benchmark.

Q58. What are the ethical concerns?

Privacy of people imaged incidentally (on-device inference, 72-hour retention, flights only in declared zones); fairness of victim detection across skin tones, body types and clothing; automation of life-critical decisions (outputs support responders, never trigger action); and dual use (audit logs, deployment policy). Our victim scores come from source-confounded, partly synthetic data, so the model must not be used operationally.

Q59. Your economic analysis claims about \$150,000 saved per year. Is that real?

It is illustrative: it assumes 100 dispatches a year, 30% avoidable false alarms and \$5,000 per dispatch; no field data. The measured cost is about 390 GPU-hours of training, roughly \$120–350 at cloud rates.

B12. Hard and curveball questions**Q60. Given all these caveats, why should we trust any of your numbers?**

Every number comes from saved logs, checkpoints and prediction files, and key ones were reproduced by re-running checkpoints (AIDER per-class results reproduce the aggregates; the D-Fire detector re-verified at 0.7711). When we found errors (HPG timing, the PANNs kernel mismatch, the audio leak) we corrected the thesis. We separate what is measured from what is claimed and give a conservative grouped estimate next to the optimistic one.

Q61. If you had three more months, what would you do first?

(1) Fix the data: synchronised RGB-IR such as FLAME 2 and real UAV audio such as DroneAudioset, and re-pair or drop ESC-50. (2) A controlled grouped rerun with the row-level recipe, three or more seeds, and leave-one-source-out evaluation. (3) Supervise the RUE with corruption labels. (4) Debug the joint detector's non-finite loss. (5) Fine-tune person detection on SARD and profile AdapFuse-v2 with INT8 on embedded hardware.

Q62. What is your single most important finding?

Near-ceiling multimodal scores on assembled public corpora can come from leakage and dataset identity rather than perception: a grouped split costs 9.60 points, a camera change costs 17–37, the victim label is decided by the source, and label-matched audio encodes the label. Any claim that fusion helps UAV perception needs synchronised sensors, degradation labels and source-decorrelated splits.

Q63. Is 98.74% versus 98.85% a meaningful difference?

No. It is 0.11 points between two single-seed runs, smaller than the run-to-run variation the no-nuisance re-run shows. The correct reading is "fusion is no better than RGB-only here", not "RGB-only is better".

Q64. Wouldn't a bigger model have solved the problem?

Not the problems we found. Larger backbones were rejected on compute grounds (4–18× the budget for at most 4.8 top-1 points), and on this split RGB-only is already at ceiling (98.85–98.99% across three backbones). The limits are the data: no independent sensors, label leakage and source confounding. A bigger model would learn those shortcuts too.

Q65. Why did you choose YOLO26, YOLO11, YOLOv10 and RT-DETR for the detection benchmark?

They cover current real-time families: the newest Ultralytics nano models (YOLO26n, YOLO11n), an NMS-free design (YOLOv10n), a transformer detector (RT-DETRv2-R18) and a larger model (YOLO26s) as a ceiling. YOLO26n is the natural HPG base: the newest nano model in the set, with NMS-free inference suited to edge use.

Part C · Cheat sheet

C1. Numbers to memorise

Item	Number	Item	Number
Corpus rows / sources	72,997 / 6	FLAME share	47,992 (65.7%)
Row-level split	51,097 / 10,949 / 10,951	Balanced training set	49,276 (12,319 × 4; 48.1% dup.)
Grouped split	50,845 / 11,055 / 11,097	Capture groups	9,267 / 2,003 / 2,002
Audio rows / unique clips	39,003 / 2,000	Test rows with audio	52.8%
AdapFuse-v1 combined F1	98.74%	RGB-only A1	98.85%
AdapFuse-v1 dis. F1 / acc.	99.13 / 99.44%	Victim F1	98.36%
Best model (EffNet-B0 RGB)	99.24%	MobileViT-XXS RGB	98.99%
Grouped AdapFuse-v1	89.14% (−9.60)	Disaster / victim drop	−14.28 / −4.93
Audio-only A3 dis. F1 / acc.	38.2 / 71.69%	Early fusion B	96.93%
UniAdapFuse v1 → v2 dis. acc.	53.71 → 99.26%	v2 detection mAP ₅₀	0.0
Best D-Fire (YOLO26s CLAHE)	77.10% (84.85 / 69.36)	YOLO26n vs HPG	71.47 vs 70.53
HPG vs baseline time	8.65 h vs 4.45 h	RT-DETRv2-R18 / YOLO11n	73.80 / 72.25
AIDER (EffNet-B0)	94.00 macro-F1 vs 88.5	FLAME (EffNet-B0)	83.72 acc. (best pub. 88.05)
Smoke severity 5	99.13 → 89.73	Gates (row-level)	RGB ≈1, th ≈1, audio ≈0.011
Params v1 / v2	12.94M / 12.30M (−4.9%)	Latency (RTX 5090)	14.32 ms, 69.9 FPS
ECE / FPR	0.0337 / 0.73%	Training budget	≈390 GPU-hours

C2. Wording to use and to avoid

Say	Don't say
"pseudo-thermal map derived from RGB"	"thermal camera", "sees heat", "night-time fallback"
"label-matched ESC-50 audio; the clip class leaks the fire/clean label"	"audio is uninformative", "synchronised audio"
"fusion matches but does not beat RGB-only on this data"	"fusion improves accuracy"
"the RUE learns run-dependent on/off weights"	"the RUE detects degraded sensors"
"HPG lowers mAP ₅₀ by 0.94 and trains ~1.9× slower"	"HPG is 2.27× faster"
"CNN6-style encoder, effectively trained from scratch"	"PANNs AudioSet transfer"
"victim-presence classification"	"survivor detection" or "person localisation"
"14.32 ms on a workstation GPU; embedded profiling is future work"	"real-time on a UAV"
"exceeds published AIDER results under Lee et al.'s protocol; below recent FLAME and D-Fire methods"	"state of the art"
"classification recovered under a revised recipe; joint detection not achieved"	"stop-gradient solved negative transfer"
"official D-Fire test split (4,306 images); published values as reported"	"we re-evaluated all published methods on our split"

C3. Glossary

Term	Meaning
RUE	Reliability and Uncertainty Estimator: MLP that outputs a gate $r \in [0, 1]$ per modality
HPG	Hazard Prior Gate: physics cue maps $\rightarrow$ bounded residual correction of the detector input
TDCG / BUSG	Top-Down Context Gate (FiLM from scene vector to detection features) / Bottom-Up Spatial Guidance (box saliency to victim head)
Combined F1	Mean of disaster macro-F1 and victim macro-F1
Row-level vs grouped split	Random frames vs whole capture groups kept on one side
CLAHE	Contrast-Limited Adaptive Histogram Equalisation, applied to D-Fire images before detection
KD	Knowledge distillation: logit (temperature 4) and feature matching from a teacher
ECE	Expected calibration error: gap between confidence and accuracy
mAP_{50}	Mean average precision at IoU 0.5